

AI MENTAL HEALTH ANA – 4 (WEEK-4)

NHS Client Insight Report

Dataset Used: Suicide Registrations in England and Wales (2024)

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1. Introduction

This report presents findings from an analysis of suicide registration statistics in England and Wales. The objective was to identify crisis-related trends, examine regional variation, and provide evidence-based insights that can support NHS decision-making and mental health service planning.

Suicide statistics provide an important indicator of severe mental distress and can help stakeholders understand areas where additional crisis intervention and support services may be required. The analysis focused on regional suicide rates, year-on-year changes between 2023 and 2024, and the identification of high-risk areas.

2. Dataset Overview

The analysis utilized the Office for National Statistics (ONS) Suicide Registrations Dataset (2024 Release).

The dataset contains information relating to:

- A. Number of suicide deaths
- B. Regional suicide rates
- C. Historical suicide trends
- D. Gender-specific statistics

E. Method-of-death classifications

The dataset provides both numerical and categorical information that can be used to understand patterns in crisis-related outcomes across England and Wales.

3. Data Quality

Before conducting the analysis, the dataset was reviewed to assess quality and suitability.

Missing Values

The regional tables used for this analysis contained very few missing values.

To ensure analytical reliability:

- A. Incomplete records were reviewed and excluded where necessary.
- B. Numerical variables were validated for consistency.
- C. Regional labels were standardized to ensure accurate comparison.

Duplicate Records

No duplicate observations were identified within the regional dataset.

This indicates that each regional record represented a unique statistical observation.

Data Formatting

Several preprocessing procedures were applied:

- A. Regional variables were standardized.
- B. National totals were removed from regional analysis.
- C. Year-on-year change indicators were calculated.
- D. Risk categories were generated to support stakeholder interpretation.

These steps ensured consistency and improved the quality of the analytical outputs.

#	Column	Non-Null Count	Dtype
0	Sex	33 non-null	object
1	Area code		
[note 2]		33 non-null	object
2	Area of usual residence		
[note 2]		33 non-null	object
3	2024		
Number of deaths		33 non-null	int64
4	2024		
Rate per 100,000			
[note 4]		33 non-null	float64
5	2024		
LCL			
[note 6]		33 non-null	float64
6	2024		
UCL			
[note 6]		33 non-null	float64
7	2023		
Number of deaths		33 non-null	int64

4. Regional Crisis Trend Analysis

The primary objective of the analysis was to understand geographical differences in crisis-related outcomes.

	Region	Rate_2023	Rate_2024	Change	Risk_Level
10	Wales	14.4	16.2	1.8	High Risk
1	North East	15.9	16.0	0.1	High Risk
2	North West	16.0	14.8	-1.2	High Risk
3	Yorkshire and The Humber	12.8	14.4	1.6	High Risk
4	East Midlands	12.6	12.9	0.3	Medium Risk
	Region	Rate_2023	Rate_2024	Change	Risk_Level
7	London	7.8	9.0	1.2	Low Risk
6	East of England	11.7	10.1	-1.6	Medium Risk
5	West Midlands	11.1	10.8	-0.3	Medium Risk
8	South East	10.9	10.9	0.0	Medium Risk
9	South West	12.6	11.8	-0.8	Medium Risk

High-Risk Regions

The highest suicide rates in 2024 were observed in:

- A. Wales
- B. North East England
- C. North West England
- D. Yorkshire and Humber

These regions consistently reported higher suicide rates than the national average.

The findings suggest that mental health challenges and crisis burden remain concentrated within specific geographical areas.

Stakeholder Implication

Regions with persistently elevated rates may require increased investment in crisis support services, suicide prevention programs, and community mental health initiatives.

Lower-Risk Regions

Several regions recorded comparatively lower suicide rates.

These included:

- A. London
- B. East of England
- C. West Midlands

Although these regions demonstrated lower overall rates, local variation may still exist within individual communities.

Stakeholder Implication

Monitoring should continue across all regions to ensure emerging risks are identified at an early stage.

5. Year-on-Year Trend Analysis

A comparison of suicide rates between 2023 and 2024 was undertaken to identify changes in crisis risk.

The analysis revealed:

- A. Several regions experienced increases in suicide rates.
- B. Some regions demonstrated relatively stable performance.
- C. Regional inequalities remained evident across both years.

The trend analysis indicates that crisis-related pressures continue to affect many areas despite ongoing mental health interventions.

	Region	Rate_2023	Rate_2024	Change
1	North East	15.9	16.0	0.1
2	North West	16.0	14.8	-1.2
3	Yorkshire and The Humber	12.8	14.4	1.6
4	East Midlands	12.6	12.9	0.3
5	West Midlands	11.1	10.8	-0.3

Stakeholder Implication

Ongoing monitoring of regional performance is essential to evaluate the effectiveness of existing support services and prevention strategies.

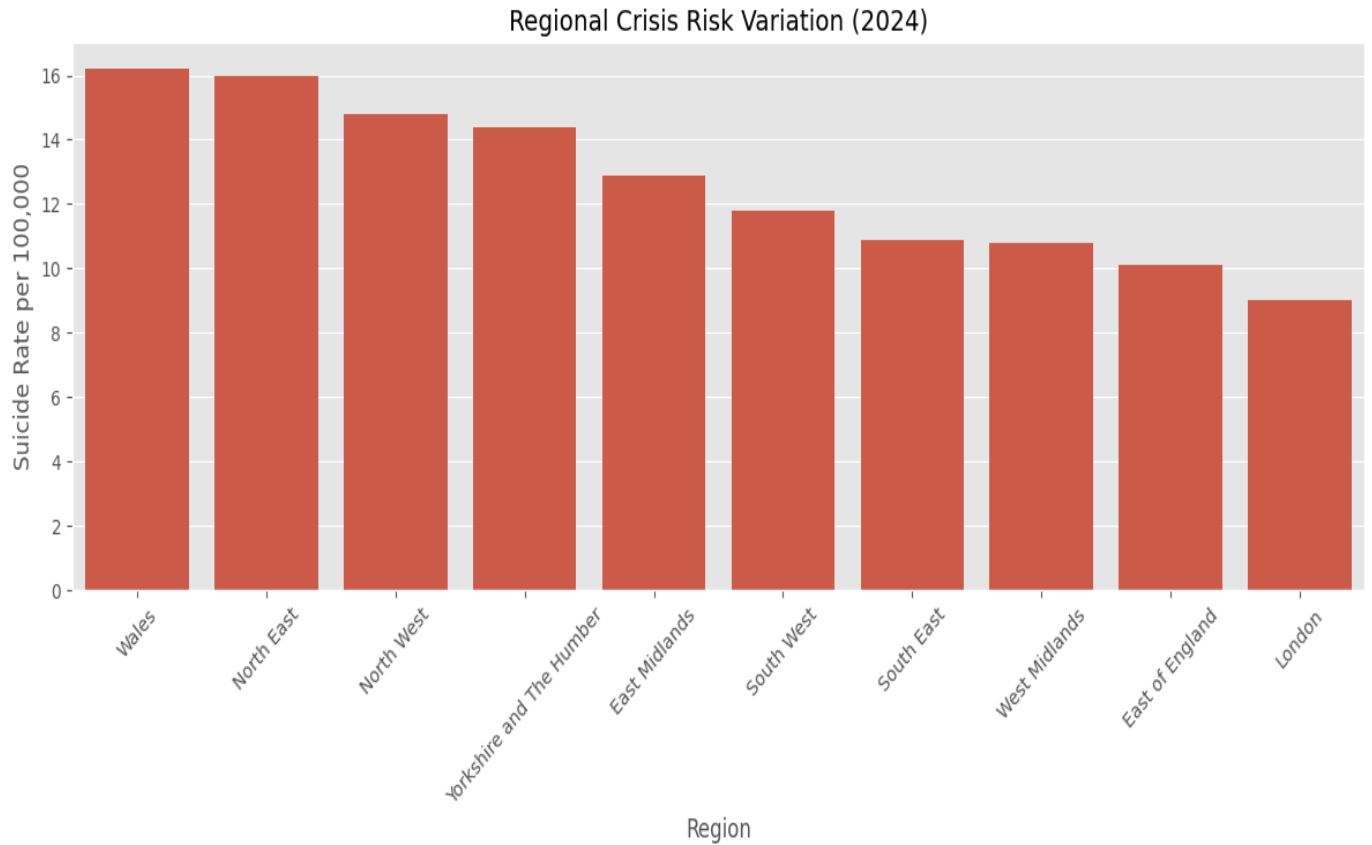
6. Data Visualization Findings

Several visualizations were developed to support interpretation and communication of findings.

Regional Crisis Risk Chart

The regional comparison chart demonstrated significant variation in suicide rates across England and Wales.

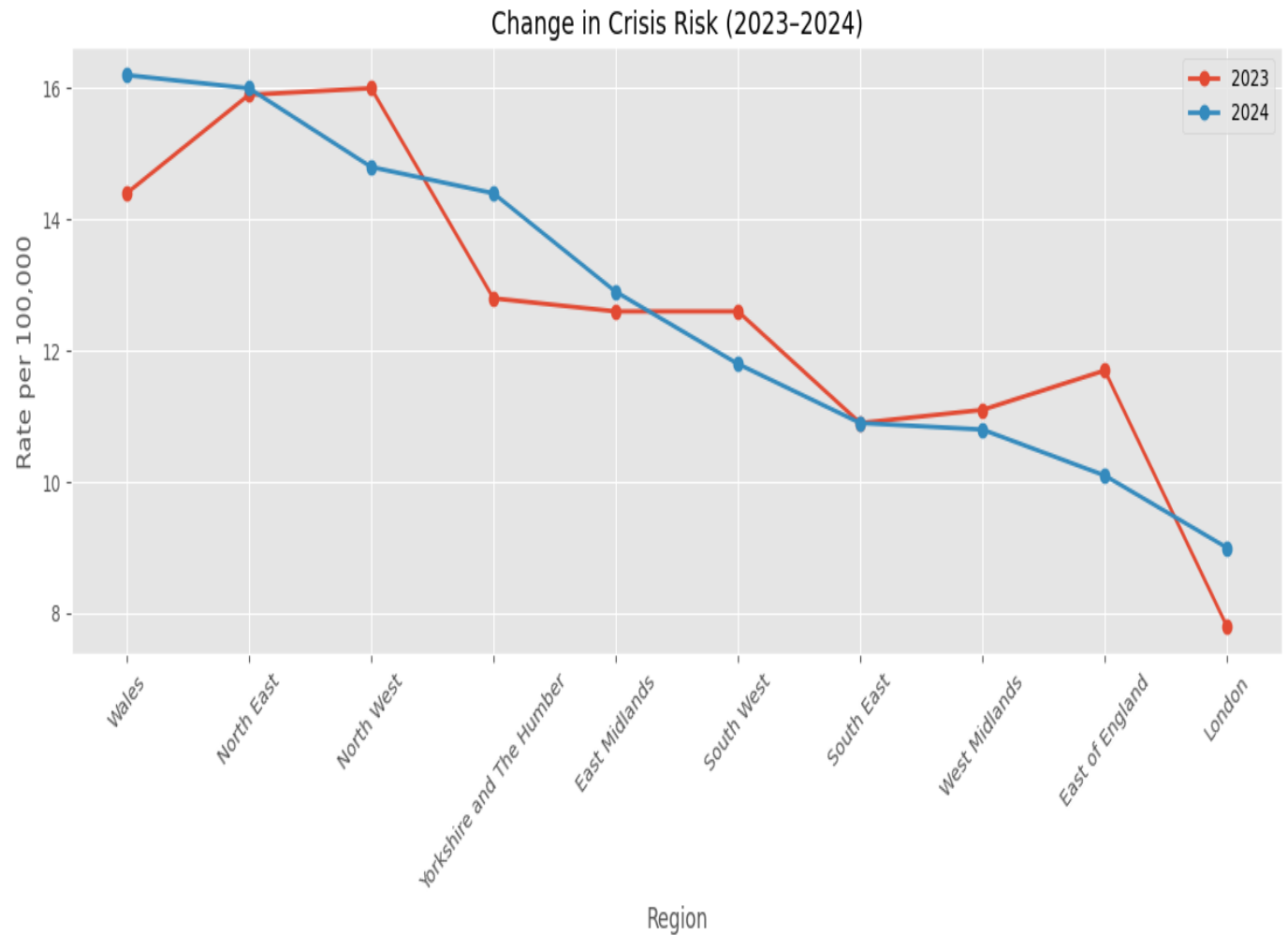
Higher-risk regions were clearly identifiable and provide a useful basis for service prioritization.



Crisis Trend Line Chart

The line chart compared suicide rates between 2023 and 2024.

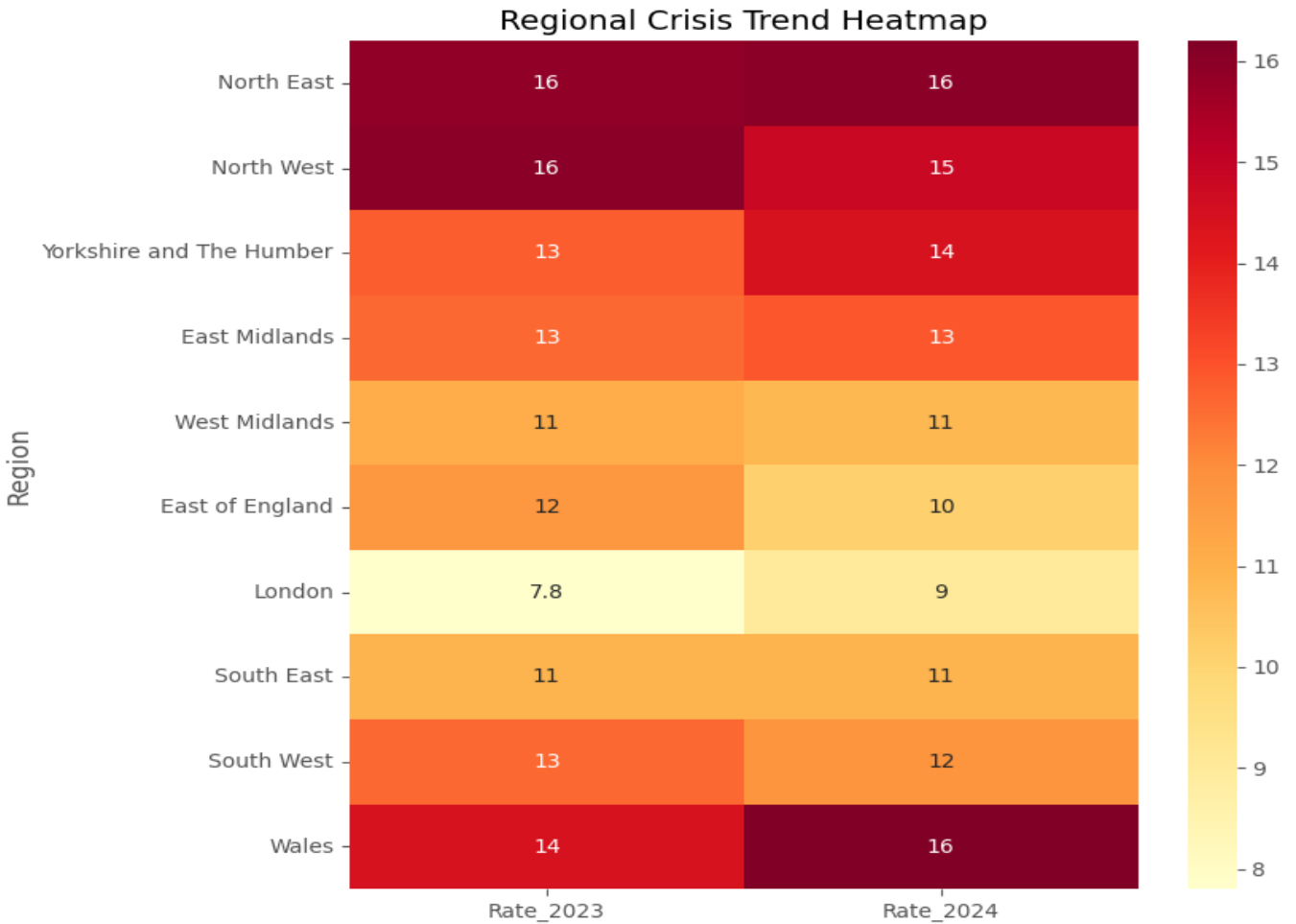
This visualization highlighted areas where crisis risk increased and enabled stakeholders to identify emerging concerns.



Regional Heatmap

The heatmap provided a concise overview of crisis intensity across regions and allowed rapid identification of high-risk areas.

The visualization supports strategic planning and resource allocation decisions.



7. Key Stakeholder Insights

Several important insights emerged from the analysis.

- A. Suicide rates continue to vary substantially across regions.
- B. Wales and several northern regions remain among the highest-risk areas.
- C. Regional inequalities continue to influence mental health outcomes.
- D. Some regions experienced worsening trends between 2023 and 2024.
- E. Crisis-support demand is likely to remain elevated within high-risk regions.
- F. Data visualization techniques provide effective tools for communicating mental health trends to decision-makers.

These findings can assist NHS stakeholders in identifying priority areas for intervention and monitoring.

8. Limitations

Several limitations should be considered when interpreting the findings.

- A. Suicide statistics represent only one measure of crisis burden.
- B. Direct indicators such as Samaritans call volumes and NHS crisis-line activity were not available within the dataset.
- C. Registration delays may influence the most recent annual figures.
- D. Regional statistics may mask important variation at local authority level.
- E. Socioeconomic and demographic variables were not included in this analysis.

Future studies would benefit from integrating additional data sources to provide a more comprehensive understanding of crisis demand.

9. Recommendations

Based on the findings, the following recommendations are proposed.

- A. Prioritize intervention efforts in regions demonstrating persistently high suicide rates.
- B. Expand crisis-support capacity in areas showing increasing risk.
- C. Strengthen early intervention and community mental health services.
- D. Improve integration of crisis-related datasets across NHS systems.
- E. Monitor regional trends on an annual basis to support evidence-based planning.

F. Incorporate demographic, socioeconomic, and healthcare access variables into future analyses.

Implementation of these recommendations may improve targeting of resources and strengthen mental health service delivery.

10. Conclusion

The analysis successfully identified regional differences in crisis-related outcomes across England and Wales and highlighted areas requiring continued attention from NHS stakeholders.

Wales and several northern English regions demonstrated the highest suicide rates, while London and parts of southern England reported comparatively lower levels of risk. Year-on-year comparisons revealed evidence of continuing crisis pressures in several regions, emphasizing the importance of sustained investment in prevention and intervention programs.

Overall, the findings provide valuable evidence to support NHS service planning, resource allocation, and future monitoring of mental health crisis trends. Continued use of data-driven approaches will help ensure that support services are targeted effectively and that emerging risks are identified at an early stage.